DATA LOADING AND DATA STORAGE

```
import pandas as pd
```

```
import os
print(os.getcwd())
print(os.listdir())
```

```
/content
['.config', 'sample_data']
```

```
# Step 1: Write 5 employee records to the file
with open('f.txt', 'w') as f:
    f.write("EmpID,Name,Department,City,Salary\n")  # Header
    f.write("101,Alice,HR,New York,75000\n")
    f.write("102,Bob,IT,Chicago,85000\n")
    f.write("103,Charlie,Finance,Houston,90000\n")
    f.write("104,Diana,Marketing,Los Angeles,78000\n")
    f.write("105,Evan,IT,Seattle,95000\n")

# Step 2: Read and display the file content
with open('f.txt', 'r') as f:
    content = f.read()

print(content)
```

```
EmpID,Name,Department,City,Salary
101,Alice,HR,New York,75000
102,Bob,IT,Chicago,85000
103,Charlie,Finance,Houston,90000
104,Diana,Marketing,Los Angeles,78000
105,Evan,IT,Seattle,95000
```

```
import pandas as pd

# Read the existing text file into a DataFrame
df = pd.read_csv('f.txt')

# Save as a CSV file
df.to_csv('f.csv', index=False)

# Display the output
print(df)
```

```
   EmpID     Name Department         City  Salary
0    101    Alice         HR     New York   75000
1    102      Bob         IT      Chicago   85000
2    103  Charlie    Finance      Houston   90000
3    104    Diana  Marketing  Los Angeles   78000
4    105     Evan         IT      Seattle   95000
```

```
# Step 1: Install openpyxl (if not already installed)
!pip install openpyxl -q

import pandas as pd

# Step 2: Read your CSV/text file
df = pd.read_csv('f.csv')  # or pd.read_csv('f.txt')

display(df)

# Step 3: Export to Excel format (.xlsx)
df.to_excel('f.xlsx', index=False)

print("Saved to f.xlsx successfully!")
```

	EmpID	Name	Department	City	Salary
0	101	Alice	HR	New York	75000
1	102	Bob	IT	Chicago	85000
2	103	Charlie	Finance	Houston	90000
3	104	Diana	Marketing	Los Angeles	78000
4	105	Evan	IT	Seattle	95000

```
Saved to f.xlsx successfully!
```

```
print(df.shape)
```

```
(5, 5)
```

```
print(df.columns)
print(df.dtypes)
df.head
```

```
Index(['EmpID', 'Name', 'Department', 'City', 'Salary'], dtype='object')
EmpID          int64
Name          object
Department    object
City          object
Salary         int64
dtype: object
```

```
pandas.core.generic.NDFrame.head
def head(n: int=5) -> Self
```

```
Return the first `n` rows.

This function returns the first `n` rows for the object based
on position. It is useful for quickly testing if your object
has the right type of data in it.
```

```
df = pd.read_csv('f.txt', sep=',')
df = pd.read_csv('f.txt', sep='\t')
```

```
df = pd.read_csv('f.csv', header=0)
df = pd.read_csv('f.csv', header=None)
```

```
df = pd.read_csv('f.csv', index_col='EmpID')
# or
df = pd.read_csv('f.csv', index_col=0)
```

```
df = pd.read_csv('f.csv', names=['ID', 'Employee_Name', 'Dept', 'Location', 'Pay'], header=0)
```

```
df = pd.read_csv('f.csv', usecols=['Name', 'Salary'])
# or
df = pd.read_csv('f.csv', usecols=[0, 4])
```

```
df = pd.read_csv('f.csv', nrows=3)
```

```
import pandas as pd


df = pd.read_csv(
    'f.csv',
    sep=',',
    header=0,
    usecols=['Name', 'Salary'],
    nrows=2
)

print(df)
```

```
    Name  Salary
0  Alice   75000
1    Bob   85000
```

DATA LOADING AND DATA STORAGE

```
#CREATE TEXT FILE ,DATA ENTER | WRITE OPERATION
with open('f.txt', 'w') as f:
```

```
    f.write("EmpID,Name,Department,City,Salary\n")  # Header
    f.write("101,RAMA,HR,hatras,75000\n")
    f.write("102,aarav,IT,agra,85030\n")
    f.write("103,priya patel,Finance,agra,90000\n")
    f.write("104,ananya,Marketing,punjab,78000\n")
    f.write("105,prince,IT,lucknow,95000")
```

```
with open("f.txt","r") as f:
  data=f.read()
  print(data)
```

```
EmpID,Name,Department,City,Salary
101,RAMA,HR,hatras,75000
102,aarav,IT,agra,85030
103,priya patel,Finance,agra,90000
104,ananya,Marketing,punjab,78000
105,prince,IT,lucknow,95000
```

SAVE THE .TXT DATA INTO CSV FILE **bold text**

```
import pandas as pd

df = pd.read_csv('f.txt')

df.to_csv('f.csv', index=False)
```

```
print(df)
```

```
   EmpID         Name Department     City  Salary
0    101         RAMA         HR   hatras   75000
1    102        aarav         IT     agra   85030
2    103  priya patel    Finance     agra   90000
3    104       ananya  Marketing   punjab   78000
4    105       prince         IT  lucknow   95000
```

```
#JSON FILE
#EXCEL
#PICKEL
df=pd.DataFrame({
    "Name": ["AMAN","AJAY","RAKESH","GEETA"],
    "MARKS": [90,98,78,76]

})
```

```
df_pickel = df
df_pickel.to_pickle("df.pkl")
```

```
st=pd.read_pickle("df.pkl")
print(st)
```

```
     Name  MARKS
0    AMAN     90
1    AJAY     98
2  RAKESH     78
3   GEETA     76
```

```
#WEB SCRAPPING
import requests
from bs4 import BeautifulSoup
url="https://www.flipkart.com/"
r=requests.get(url)
soup=BeautifulSoup(r.text,"html.parser")
```

```
print(soup.title.text)
```

```
Online Shopping Site for Mobiles, Electronics, Furniture, Grocery, Lifestyle, Books & More. Best Offers!
```

```
url="https://www.amazon.com/"
r=requests.get(url)
soup=BeautifulSoup(r.text,"html.parser")
print(soup.title.text)
```

```
Amazon.com. Spend less. Smile more.
```

```
#parquet
df_perquet=df_pickel
```

```
print(df_perquet)
```

```
     Name  MARKS
0    AMAN     90
1    AJAY     98
2  RAKESH     78
3   GEETA     76
```

```
df_perquet.to_parquet("df.parquet",index=False)
```

```
df_perquet_file=pd.read_parquet("df.parquet")
```

```
print(df_perquet)
```

```
     Name  MARKS
0    AMAN     90
1    AJAY     98
2  RAKESH     78
3   GEETA     76
```

Sqlite

```
import sqlite3
conn=sqlite3.connect("students.db")
print("Opened database successfully")
```

```
Opened database successfully
```

```
cursor=conn.cursor()
```

```
cursor.execute("""
CREATE TABLE IF NOT EXISTS students_table(
NAME TEXT,
ROLL_NO INTEGER,
MARKS INTEGER
)
""")
print("Table created successfully")
```

```
Table created successfully
```

```
import sqlite3
# Ensure we have an active connection and cursor
conn = sqlite3.connect("students.db")
cursor = conn.cursor()

# write the data
cursor.execute("INSERT INTO students_table VALUES('AMAN',1,90)")
cursor.execute("INSERT INTO students_table VALUES('AJAY',2,98)")
cursor.execute("INSERT INTO students_table VALUES('RAKESH',3,78)")
cursor.execute("INSERT INTO students_table VALUES('GEETA',4,76)")
conn.commit()
```

```
print("Records inserted successfully")
conn.close()
```

```
Records inserted successfully
```

```
#Retreve | read the data
conn=sqlite3.connect("students.db")
cursor=conn.cursor()
cursor.execute("SELECT * FROM students_table")
row=cursor.fetchall()
for row in row:
  print(row)
```

```
('AMAN', 101, 98)
('AJAY', 102, 99)
```

```
('AMAN', 1, 90)
('AJAY', 2, 98)
('RAKESH', 3, 78)
('GEETA', 4, 76)
```